

A Positive Minimal Laplacian Solves the Sectorial Numerical-Range Question

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Status

`candidate_full_solution_likely_valid`. This note gives an affirmative example for the final open question of Adler–Dada–Radl [1].

Source question

For a closed densely defined operator A on a complex Banach space, the source defines its numerical spectrum $\sigma_n(A)$ through half-planes contained in the ordinary resolvent and satisfying the sharp Hille–Yosida resolvent estimate. It asks whether one can have

$$\overline{\operatorname{co}} W(A) \subset \Sigma_\delta^\theta + z \quad \text{and} \quad \sigma_n(A) = \mathbb{C},$$

where $0 < \delta < \pi/2$ and $W(A)$ is the spatial numerical range [1, p. 14].

Finally we ask the following open question:

Is there an operator $(A, D(A))$ such that $\overline{\operatorname{co}}\{\langle Ax, j(x) \rangle : x \in D(A), \|x\| = 1, j(x) \in \mathfrak{J}(x)\}$ is (contained in) a sector $\Sigma_\delta^\theta + z$ and $\sigma_n(A) = \mathbb{C}$?

Idea of the proof

A positive symmetric operator automatically has numerical range on the positive real ray, hence in a proper sector. It need not be self-adjoint. If its domain imposes more boundary conditions than the order of its differential expression, the adjoint has no boundary conditions at all. The adjoint then has an eigenfunction at every complex parameter. Range–kernel orthogonality makes the range of $\lambda - A$ non-dense for every λ , so the ordinary resolvent, and therefore the numerical resolvent, is empty.

Construction and theorem

Let $H = L^2(0, 1)$ over $\mathbb{C}$, and define

$$\begin{aligned} Af &= -f'', \\ D(A) &= \{f \in H^2(0, 1) : f(0) = f'(0) = f(1) = f'(1) = 0\}. \end{aligned}$$

Theorem 1. *The operator A is closed and densely defined. Its closed convex numerical range is contained in the positive real ray, and hence in Σ_δ^0 for every $0 < \delta < \pi/2$. Nevertheless,*

$$\sigma_n(A) = \mathbb{C}.$$

Thus A answers the source question affirmatively.

Proof. The inclusion $C_c^\infty(0,1) \subset D(A)$ shows that A is densely defined. To see closedness directly, note that for $f \in D(A)$,

$$f'(x) = \int_0^x f''(t) dt, \quad f(x) = \int_0^x (x-t)f''(t) dt.$$

Consequently the graph norm controls the H^2 norm. If $f_k \rightarrow f$ and $Af_k \rightarrow h$ in L^2 , then (f_k) is Cauchy in H^2 . Its H^2 limit has the four zero traces, equals f in L^2 , and satisfies $-f'' = h$. Hence A is closed.

For $f \in D(A)$, integration by parts gives

$$\langle Af, f \rangle = \int_0^1 -f''(x)\overline{f(x)} dx = \int_0^1 |f'(x)|^2 dx \geq 0.$$

Thus $W(A) \subset [0, \infty)$, and the same is true of its closed convex hull. In particular it lies in every sector Σ_δ^0 with $0 < \delta < \pi/2$.

We next compute the adjoint. Two integrations by parts show that every $g \in H^2(0,1)$ belongs to $D(A^*)$ and

$$A^*g = -g'';$$

all boundary terms vanish because both f and f' vanish at both endpoints. Conversely, if $g \in D(A^*)$, testing against $C_c^\infty(0,1)$ shows in the distributional sense that $g'' \in L^2(0,1)$. The one-dimensional Sobolev characterization then gives $g \in H^2(0,1)$. Therefore

$$D(A^*) = H^2(0,1), \quad A^*g = -g''.$$

Fix an arbitrary $\lambda \in \mathbb{C}$. The second-order initial-value problem

$$-g'' = \bar{\lambda}g, \quad g(0) = 1, \quad g'(0) = 0$$

has a nonzero smooth solution on $[0,1]$. Hence

$$0 \neq g \in \ker(\bar{\lambda}I - A^*).$$

The Hilbert-space range–kernel identity yields

$$\text{Ran}(\lambda I - A)^\perp = \ker(\bar{\lambda}I - A^*) \neq \{0\}.$$

Thus $\text{Ran}(\lambda I - A)$ is not dense for every $\lambda \in \mathbb{C}$. In fact, $\lambda I - A$ is injective as well: a solution of the homogeneous ODE with $f(0) = f'(0) = 0$ is identically zero. Therefore every λ belongs to the residual spectrum in both the source convention and the usual injective-range convention.

It follows that the ordinary resolvent $\rho(A)$ is empty. The numerical resolvent $\rho_n(A)$ is, by definition, a union of half-planes contained in $\rho(A)$, so it too is empty. Consequently

$$\sigma_n(A) = \mathbb{C} \setminus \rho_n(A) = \mathbb{C}.$$

This proves all assertions. □

Verification notes

No computation is used. The proof can be checked through four independent standard identities:

1. $C_c^\infty(0, 1)$ gives density, and the two integral formulas give graph closedness.
2. Green's formula gives positivity and the sector inclusion.
3. The four minimal boundary conditions make the adjoint domain the unrestricted maximal domain $H^2(0, 1)$.
4. $\text{Ran}(\lambda - A)^\perp = \ker(\bar{\lambda} - A^*)$ converts the two-dimensional ODE solution space into non-dense range for every parameter.

Novelty check and limitations

The lightweight run indexes and local corpus were searched for arXiv:1507.01418, the exact final question, and combinations of “numerical spectrum”, “sector”, “residual spectrum”, and “minimal Laplacian”. Bounded web and arXiv searches with the same terms found the source and later papers on related essential numerical spectra, but no explicit later answer to this question. The construction is nevertheless based on classical minimal symmetric-operator facts, so the novelty confidence is moderate rather than definitive.

The example is Hilbertian and intentionally has empty ordinary resolvent. This is fully allowed by the source question. It does not address a stronger variant requiring nonempty ordinary resolvent or generation by an unrotated semigroup.

Human review recommendation

Check the adjoint-domain computation and the source convention for residual spectrum. Once $D(A^*) = H^2(0, 1)$ is confirmed, the conclusion follows immediately for every complex λ .

References

- [1] M. Adler, W. Dada, and A. Radl, *A semigroup approach to the numerical range of operators on Banach spaces*, arXiv:1507.01418 (2015); Semigroup Forum **94** (2017), 51–70.